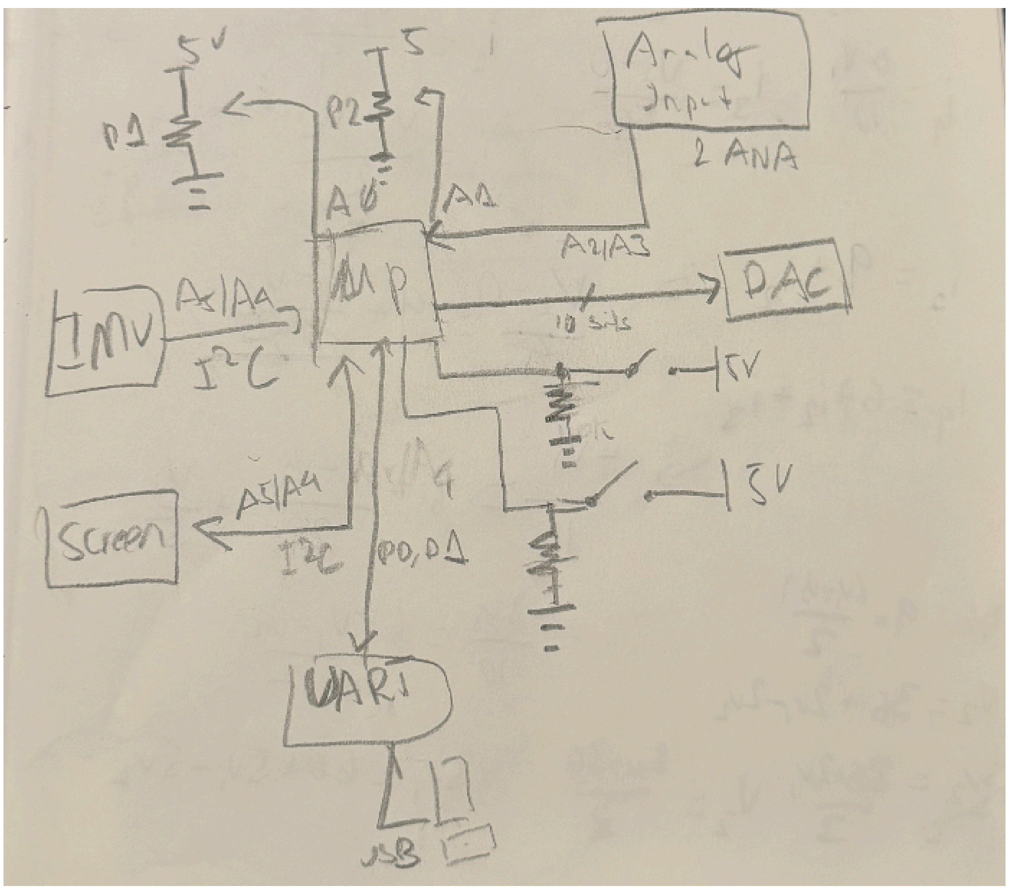
Introduction

In this project, we were tasked with designing and building an Arduino Uno shield. We especially made a DSP device with a 10-bit DAC and used KiCad for the PCB design. The purpose of this project was to familiarize oneself with the tools, methodologies, and general process of building and designing microprocessors.

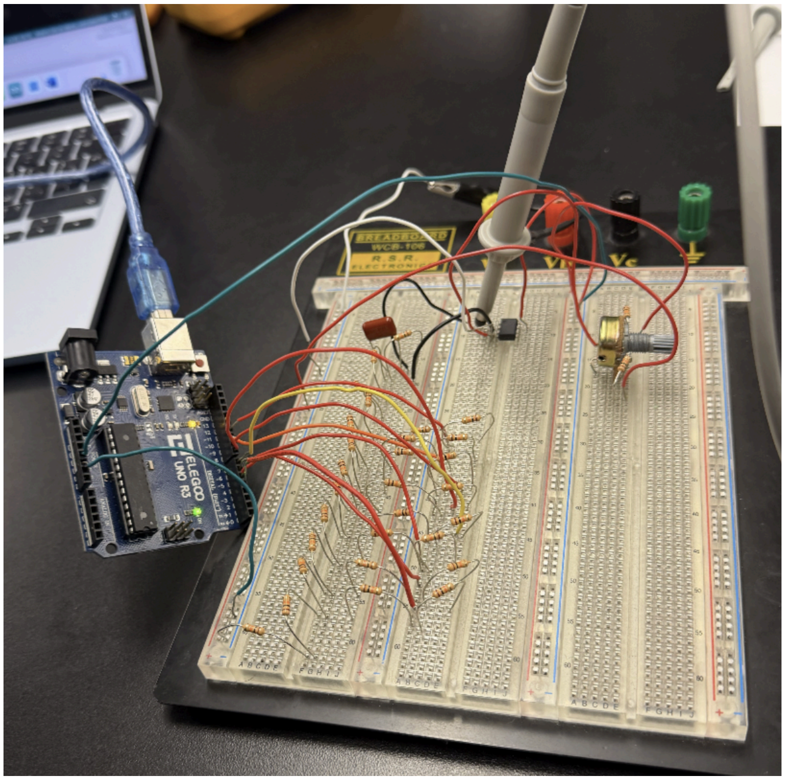
Design Stage

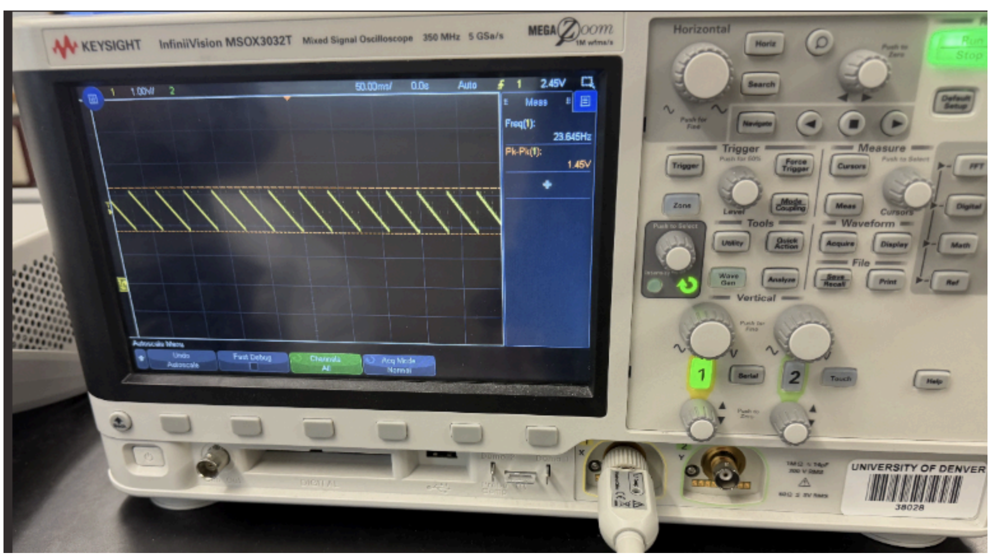
To start the process, we designed a block diagram for the shield, consisting of a 10bit DAC, OLED screen, IMU, Analog Input, UART, three potentiometers, two analog inputs, and two buttons alongside the main microcontroller. At this stage, we clearly defined what components would be involved and how they'd interact with each other in this system.



Prototype Stage

Before fully designing the shield, we first constructed an R2R ladder DAC. This was done on a breadboard, and we were also assigned to write a program to display output on an oscilloscope based on our DAC.



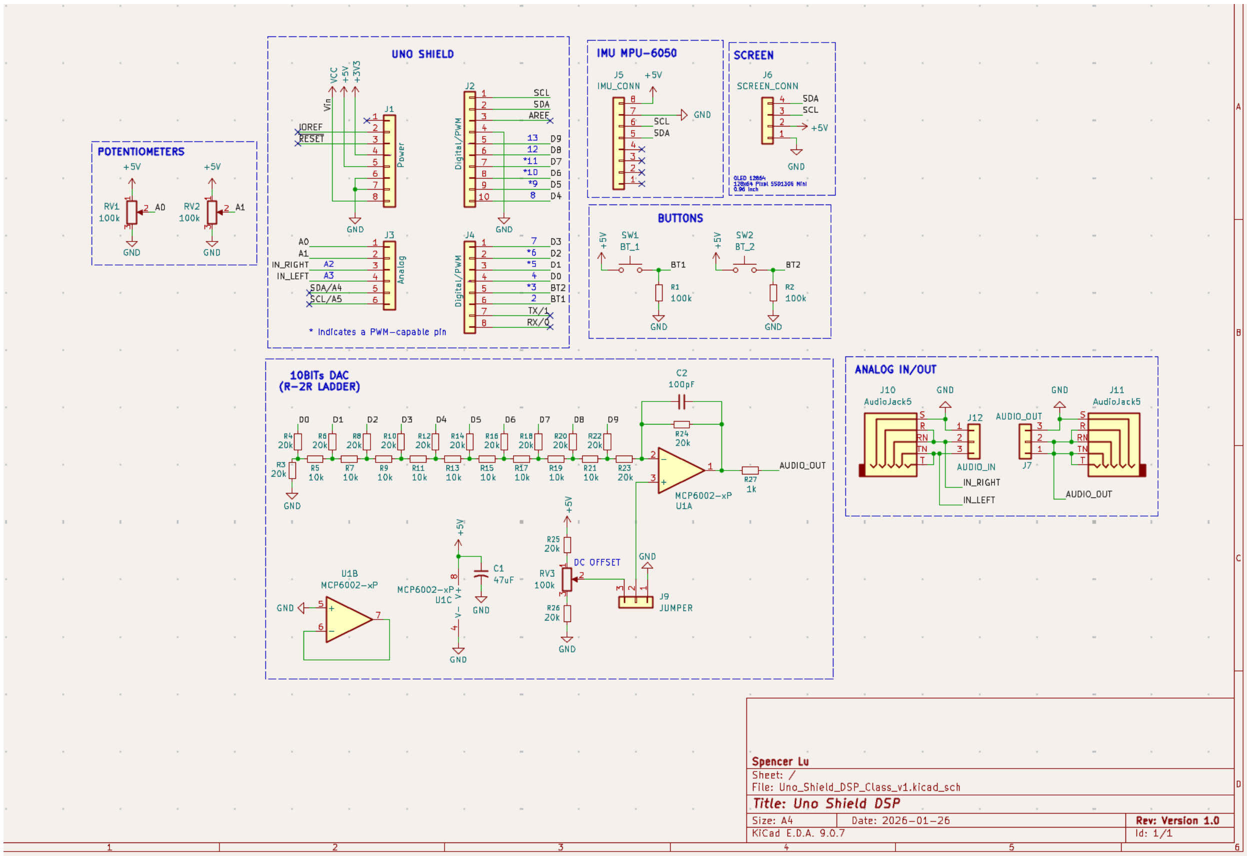


```
/*
Scope test for Phase B prototype:
- 10-bit R-2R DAC data bus: D4..D13
- Outputs a repeating ramp 0..1023 to the R-2R ladder
- Also outputs a 1 kHz square wave on D3 for quick scope sanity check

How to use:
1) Probe square wave: scope tip on D3, ground on Arduino GND -> should see 1 kHz square wave.
2) Probe DAC output: scope tip on the analog output of your R-2R ladder (or op-amp output),
ground on Arduino GND -> should see a ramp/staircase waveform.
*/

#include <Arduino.h>

const uint8_t SQUARE_PIN = 3; // quick test output
const uint8_t dacPins[10] = {4,5,6,7,8,9,10,11,12,13}; // 10-bit bus

// Write 10-bit value (0..1023) to R-2R ladder
void writeDAC10(uint16_t value)
{
value &= 0x03FF; // keep only 10 bits

// D4 = bit0 (LSB), D13 = bit9 (MSB)
for (int bit = 0; bit < 10; bit++)
{
digitalWrite(dacPins[bit], (value >> bit) & 1);
}
}

void setup()
{
pinMode(SQUARE_PIN, OUTPUT);
digitalWrite(SQUARE_PIN, LOW);

for (int i = 0; i < 10; i++)
{
pinMode(dacPins[i], OUTPUT);
digitalWrite(dacPins[i], LOW);
}
}

void loop()
{
// ----- 1 kHz square wave on D3 (sanity check) -----
// 1 kHz period = 1000 us, half period = 500 us
digitalWrite(SQUARE_PIN, HIGH);
delayMicroseconds(500);
digitalWrite(SQUARE_PIN, LOW);
delayMicroseconds(500);

// ----- DAC ramp 0..1023 -----
// One ramp step per loop call would be too slow because of the square wave delays,
// so we do a short burst of DAC steps each time.
for (uint16_t code = 0; code < 1024; code++)
{
writeDAC10(code);

// Controls ramp speed:
// - smaller = faster (higher output frequency)
// - bigger = slower (easier to see stair-steps)
delayMicroseconds(80);
}
}
```

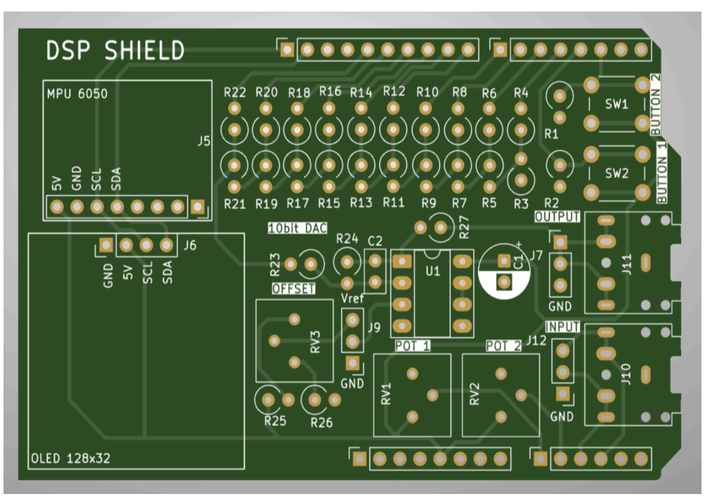
PCB Design

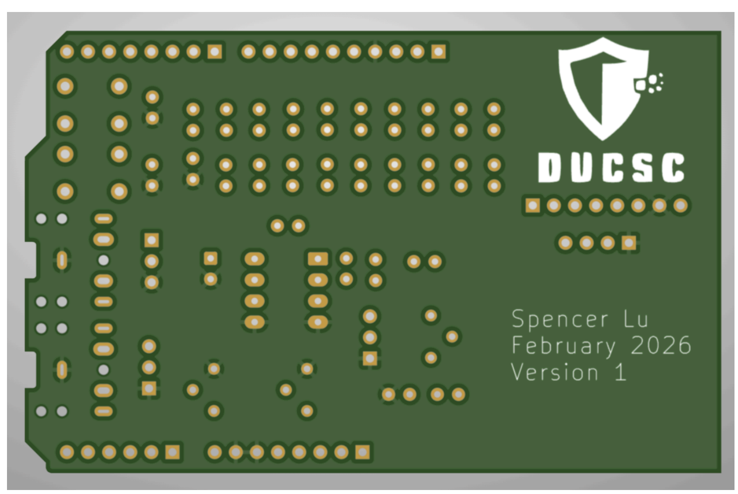
After the initial prototype, we then constructed the schematics, layout, gerber files, and 3D render in KiCad.

Schematics:

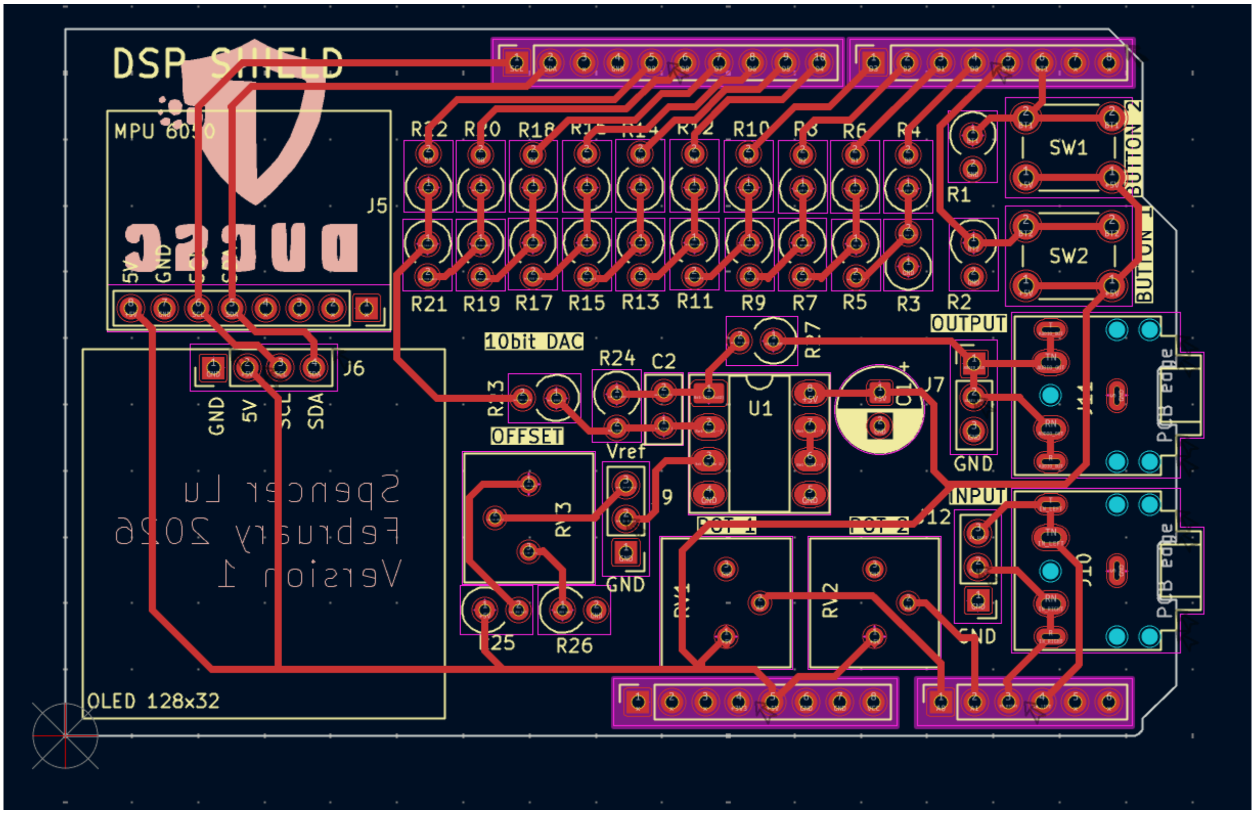


Gerber front and back view:

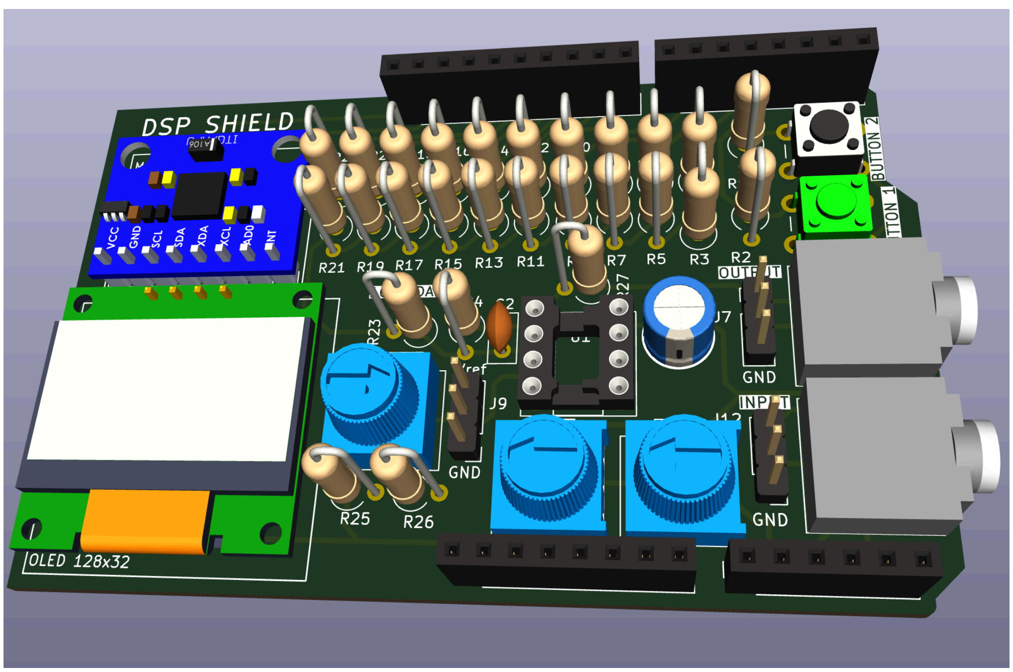


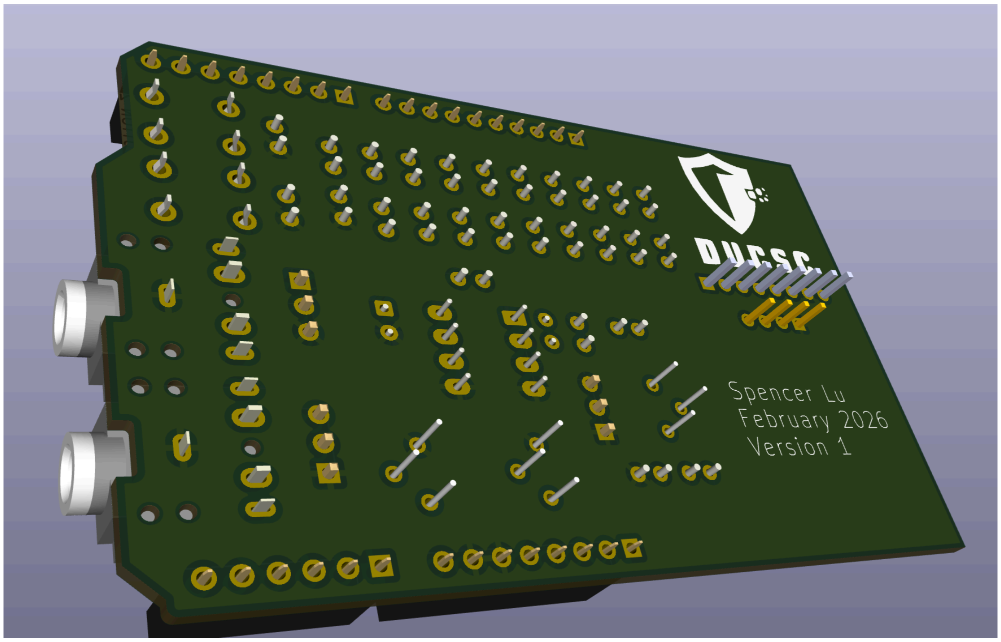


Fully finished Design in KiCad:

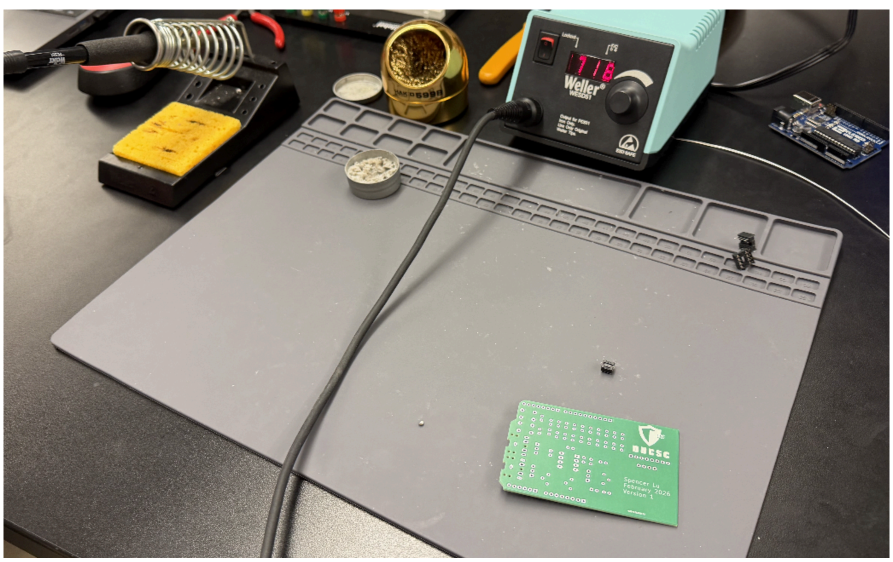


Front and back 3D render:

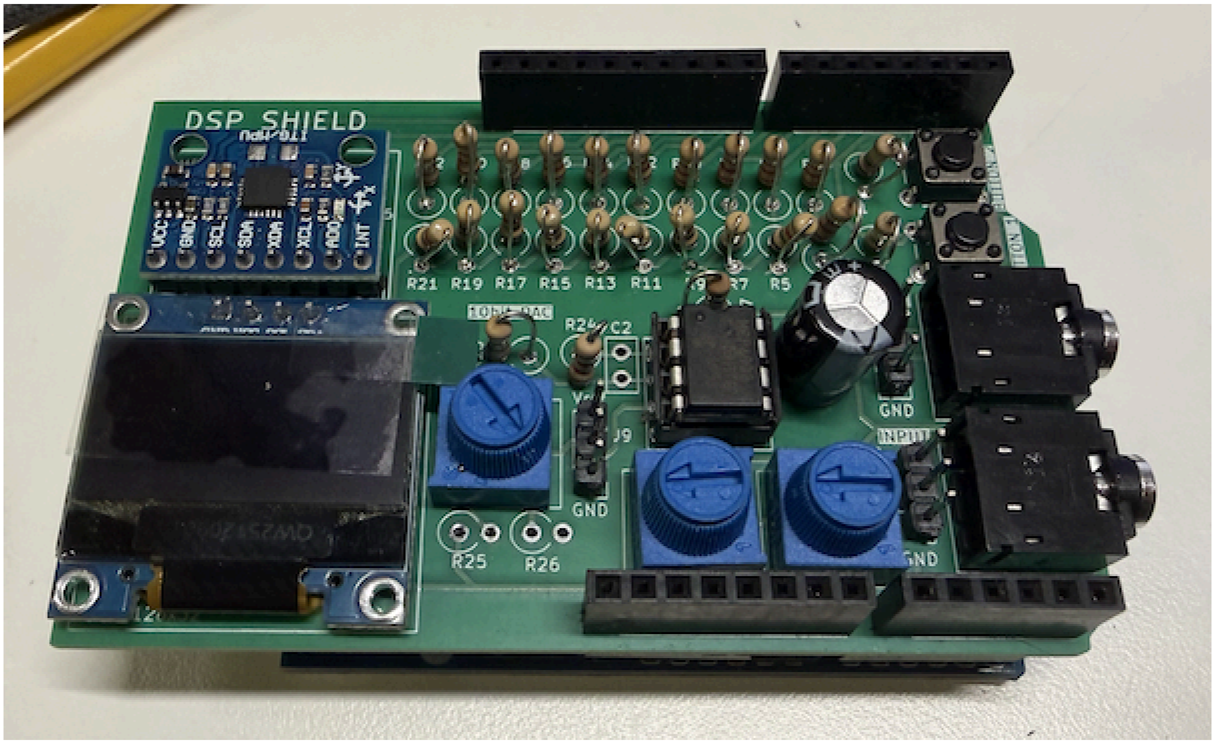




<u>Assemble Stage</u>
We then assembled the boards by soldering each pieces individually.

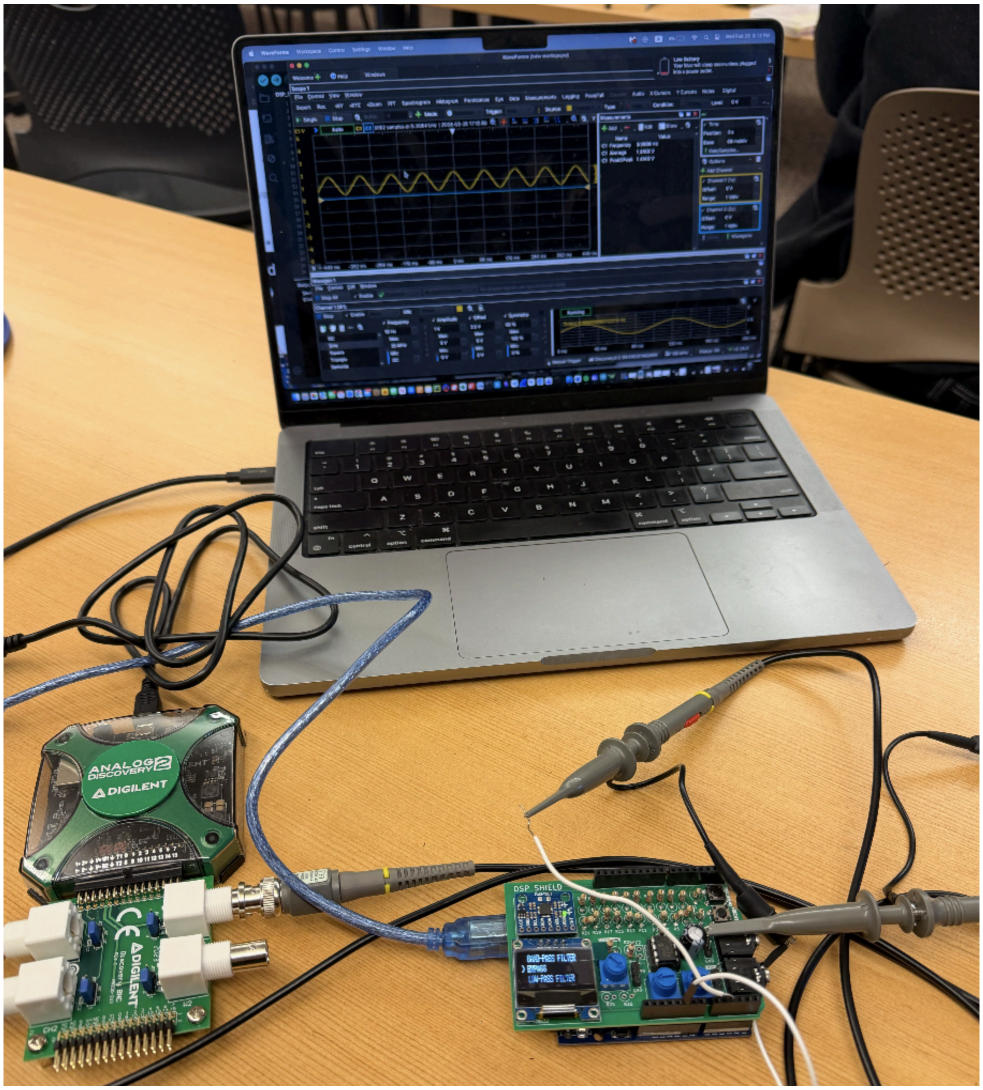


Final product:



Testing

Finally, to test the functionality of our microprocessor, our instructor provided test code for use with an oscilloscope to view output and confirm it worked. The code for this can be found in the Test Code directory.



In this snippet, my DSP Shield successfully ran the shield filters test code successfully.